



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering

Academic Year 2023 – 2024 (Even Semester)

Degree, Semester & Branch: II Semester B.E. CSE-'B'

Course Code & Title: CS3251 Programming in C

Name of the Faculty member (s): Mrs.S.Manjula

Innovative Practice Description

Unit / Topic: Unit II / String Operations

Course Outcome: CO 2

Topic Learning Outcome: TLO 6

Activity Chosen: Think Pair Share

Justification:

- The purpose of the activity is to evaluate the understanding of every student on an individual basis by promoting peer knowledge sharing and independent thinking. Programming requires an understanding of string principles.
- In addition to offering students a chance to showcase their individual comprehension, the "think-pair-share" practice fosters collaborative learning.
- **Time Allotted for the Activity: 15 minutes**

Details of the Implementation:

- The instructor introduced specific concepts or topics in the classroom within a 30-minute timeframe, presenting a program to the students.
- The instructor displayed the university question on a string to students.
- The question was asked in Nov/Dec 2023 and the question was

12. (a) (i) Write the string handling function used to do the following.

(2+2+2+2)

- If first name, last name and middle name are given as input, display it as full name.
- Display the number of characters in full name.
- Convert the first name into capital letters and last name into small letters.
- Print the full name in reverse order.
- Students were given 1-5 minutes to independently think of how to solve the programming question.
- After giving the students some time to work independently, the instructor instructs them to pair up with a partner. They then discuss their solutions with each other, explaining their thought processes, sharing different approaches, and providing feedback. (6-10 minutes).
- Students then shared their solutions with the entire class during the last segment of 11-15 minutes.

- The instructor finally discussed the answer.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO9	PO10	PSO1
CO 2	2	1	1	1	1	1

(1 – Low 2 – Moderate 3 – High)

PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1
	2	1	1	1	1	1
Justification for correlation	The activity included thinking and made the students able to think about problem-solving approaches.	Students were able to approach the program by applying what they learned.	A solution for the problem was provided using string functions.	As students were involved in the activity as a team, their teamwork skills improved.	Students' communication skills improved as they faced the audience and communicated the activity.	Students were able to apply C programs to provide solutions for information and communication technology-related problems.

- Images / Screenshot of the practice:**



Fig:1 Glimpse of Think Pair Share Activity



Fig:2 Glimpse of Think Pair Share Activity

• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

The students felt that by working independently first and then discussing their solutions with a partner, they were prompted to think critically about the problem and evaluate different solutions.

Benefit of the practice:

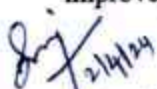
- The activity promoted communication skills as students expressed their thoughts, explained their solutions to their partners, and participated in the class-wide sharing session.
- From this activity, the students can get more clarity in the particular topic.
- This practice has made the class more interactive than the conventional practice

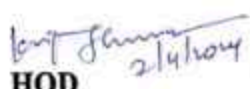
Challenges faced in implementation:

- In some pairs, one student may dominate the discussion while the other remains passive. This can hinder the effectiveness of the activity as it doesn't fully engage all students.
- Some students may feel uncomfortable sharing their thoughts with a partner or the whole class, especially if they are shy or lack confidence in their abilities.

References:

1. <https://www.ritrjpm.ac.in/images/computer-science/TPS-Venkat-GE6075.pdf>
2. <https://www.readingrockets.org/strategies/think-pair-share>
3. <https://www.wgu.edu/heyteach/article/how-think-pair-share-activity-can-improve-your-classroom-discussions1704.html>


Signature of Faculty Member


HOD